

manently include a reference capacitor, say, 100nF. With a reference capacitor connected, you can easily set the calibration resistance using preset VR3 to obtain a full-scale deflection. This would finish the calibration for other ranges as well. The capacitance of C_x is given by the following expression:

$$C_x = \frac{\text{Duty cycle} \times \left(\frac{V_{C2} - V_{C1}}{R_C + R_M} \right) \times N_D}{I_{FSD}}$$

Here, V_{C2} and V_{C1} are the output gate voltages of N4 and N2, respectively. N_D is the total number of divisions on the ammeter scale, R_M and $R_C = VR3$ are the meter and series calibration resistances, and I_{FSD} is the full-scale deflection current of the meter.

The range of the instrument can be further extended by including lower valued range resistances (1k Ω gives 0-100nF and 100 Ω gives 0-1 μ F). We have used a basic instrument to keep the diagram as simple as possible.

Since NE555 cannot supply the required current maintaining a con-

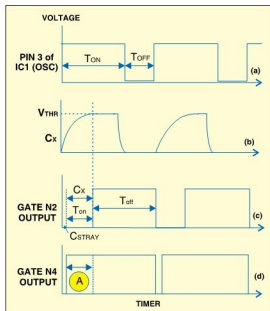


Fig. 2: Timing diagram showing different waveforms originating at different points in the circuit

stant voltage, additional driver stage of transistors T1 (2N3904) and T2 (N3906) have been included. The pulsed drive automatically cuts-off the supply during off-cycle, enabling a proper discharge of C_x , which is im-

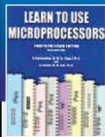
portant to get the right value of the capacitor. Otherwise a partially charged capacitor will be subjected to the next charge cycle and indicate a lower value. An indicative timing diagram (not to scale) is given in Fig. 2.

The circuit takes some time to stabilise thermally and is more suitable as a lab instrument. It is recommended to include a lower-valued variable resistance in series with the preset controlling the current through the meter. This will allow easy frequent calibration of the instrument.

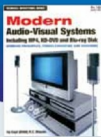
Assemble the circuit on a general-purpose PCB in a suitable enclosure. Approximately 63.5cm (25-inch) long test leads with crocodile clips are used for compensating the capacitance of $C3$. This capacitance can be changed according to the length of the test leads. Fix the ammeter in the front panel. ●

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